

B-PROFILE-MECHANISM-KILL-TEST-v1: Can profile have $\bar{\alpha} \leq 0$ everywhere?

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1 Question

Can blow-up profile compatible with $C_F^{conc} + C_I^{freq} + C_J^{neg}$ have $\bar{\alpha} = \bar{\xi}^T \bar{S} \bar{\xi} \leq 0$ everywhere? Needed for rescue: $\int_{\{\bar{\alpha} > 0\}} |\bar{\omega}|^2 \bar{\alpha} \geq c_0 > 0$.

2 Limit equations

Rescaling $u^{(n)}(y, s) = r_n u(r_n y, T^* + r_n^2 s)$, $r_n \rightarrow 0$, $u^{(n)} \rightarrow \bar{u}$. Limit satisfies ancient NS: $\partial_s \bar{u} + \bar{u} \cdot \nabla \bar{u} + \nabla \bar{p} = \nu \Delta \bar{u}$, $\nabla \cdot \bar{u} = 0$, on $(-\infty, 0]$. If $\nu r_n^{-2} \rightarrow 0$, Euler ancient. C_F^{conc} : $\|\bar{u}\|_{L^3} \lesssim 1$ or concentration measure. C_I^{freq} : $\text{supp } \hat{u}$ at high freq or tightness. C_J^{neg} : allows compressive epochs, no sign forced. Enstrophy balance for profile: $\frac{1}{2} \partial_s \|\bar{\omega}\|_2^2 + \nu \|\nabla \bar{\omega}\|_2^2 = \int \bar{\omega}^T \bar{S} \bar{\omega} = \int |\bar{\omega}|^2 \bar{\alpha}$.

3 Search for identity forcing $\bar{\alpha} > 0$

Try: $\int \bar{\omega}^T \bar{S} \bar{\omega} = \int \bar{S} : (\bar{\omega} \otimes \bar{\omega})$. Since $\text{tr } \bar{S} = 0$, $\lambda_1 + \lambda_2 + \lambda_3 = 0$. $\bar{\alpha} = \lambda_1 \cos^2 \theta_1 + \lambda_2 \cos^2 \theta_2 + \lambda_3 \cos^2 \theta_3$ where $\cos \theta_i = \bar{\xi} \cdot e_i$. If $\bar{\xi}$ aligns with most negative eigenvalue, $\bar{\alpha} \leq 0$ possible.

Is there integral identity forbidding global ≤ 0 ? Compute: $\int |\bar{\omega}|^2 \bar{\alpha} = \int \bar{\omega}^T \bar{S} \bar{\omega} = - \int \bar{u} \cdot (\bar{\omega} \cdot \nabla \bar{\omega})$? Integration by parts gives $\int \bar{S} : \bar{\omega} \otimes \bar{\omega} = - \int \bar{u} \cdot \text{div}(\bar{\omega} \otimes \bar{\omega})$. But $\text{div}(\bar{\omega} \otimes \bar{\omega}) = \bar{\omega} \cdot \nabla \bar{\omega}$, no sign.

Biotsavart: $\bar{S} = \mathcal{R}[\bar{\omega}]$ Riesz transform, nonlocal, no pointwise sign.

Candidate identities: enstrophy production can be zero for 2D flows, for shear flows $\bar{u} = (f(y), 0, 0)$, $\bar{\omega} = (0, 0, -f')$, $\bar{S}$ has zero diagonal in vorticity direction? Example: $\bar{\alpha} = 0$ identically for parallel shear, nontrivial profile, satisfies NS (steady) with $\bar{\alpha} = 0$. Example: Kolmogorov-type: $\bar{u} = (\sin y, 0, 0)$, $\bar{\omega} = (0, 0, \cos y)$, $\bar{S}_{12} = \frac{1}{2} \cos y$, $\bar{\xi} = (0, 0, 1)$, $\bar{\alpha} = 0$.

Thus $\boxed{\text{nontrivial profile with } \bar{\alpha} \equiv 0 \text{ exists as Euler/NS solution}}$, compatible with incompressibility.

With C_J^{neg} , such shear belongs to admissible class: no positive stretching yet infinite L^3 ? Shear has infinite energy but localized version can be made L^3 -bounded via cutoff, keeping $\bar{\alpha} = 0$.

Weight: even if we find point with $\bar{\alpha} > 0$, need $c_0 > 0$. Shear example shows $c_0 = 0$ compatible.

Type I rate $r(t) \sim (T^* - t)^{1/2}$ not used, but without it no control on $\int W_+$ vs profile integral. $r_n^{-1} \int |\bar{\omega}|^2 \bar{\alpha}$ could be zero even if $r_n \rightarrow 0$.

4 Check 1-5

1. Limit equations obtained, no incompressibility violation. 2. Identity $\int |\bar{\omega}|^2 \bar{\alpha}$ is enstrophy production, no sign. 3. Sign not excluded: shear gives counterexample $\bar{\alpha} = 0$. 4. Weight $c_0 > 0$ not forced. 5. C_F, C_I, C_J compatible with shear-type profile after localization.

5 Kill test

$\bar{\alpha} \leq 0$ (indeed $= 0$) compatible with limit equations and $\mathcal{C}_{\text{PROVEN}}$ constraints. No quantitative $\int_{\{\bar{\alpha} > 0\}} |\bar{\omega}|^2 \bar{\alpha} \geq c_0 > 0$ derived. Only $\bar{\alpha} > 0$ at some point would be insufficient anyway. Requires no Type I, but still fails.

$B = KILLED$	No identity profile $\Rightarrow \bar{\alpha} > 0$ with weight
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6 Conclusion

$G8-DESIGN-v2 = CLOSED/BLOCKED$	$G8-01 = G8-02 = G8-03 = KILLED, A = B = C = BLOCKED$
$\mathcal{C}_{\text{PROVEN}} \Rightarrow \perp OPEN\ intact$	$NO\ G8-04$ Class scale/profile/ $S-\Omega$ exhausted without fabrication. ZFL clean.